Proof by Contradiction

Exam code: 9MA0

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Help me understand

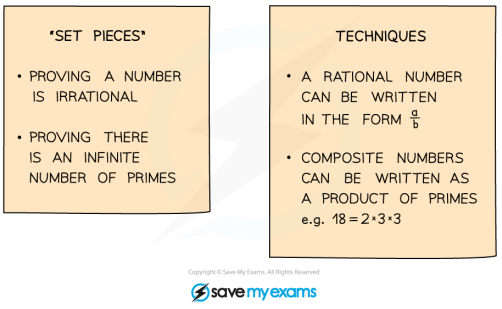
Proof by contradiction

What is proof by contradiction?

- A proof by contradiction assumes the **opposite** result is true
- Then, through a series of logical steps, shows that this cannot be so

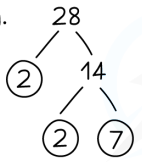
How do I use proof by contradiction?

- This type of proof often involves explanation in words alongside mathematical statements
- Start with a statement assuming the opposite is true
  - e.g. 'Assume...'



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

e.g.



$28 = 2 \times 2 \times 7 = 2^2 \times 7$

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Worked Example

Prove that $\sqrt{5}$ is irrational.

ASSUME $\sqrt{5}$ IS RATIONAL

START WITH THE OPPOSITE

SO $\sqrt{5} = \frac{a}{b}$ WHERE a AND b ARE INTEGERS WITH NO COMMON FACTORS AND $b \neq 0$

SQUARING BOTH SIDES: $5 = \frac{a^2}{b^2}$

$\Leftrightarrow 5b^2 = a^2$

THEREFORE a^2 IS A MULTIPLE OF 5
SO a IS A MULTIPLE OF 5

SINCE a IS AN INTEGER

WRITING $a = 5k$ FOR SOME INTEGER k,

$a^2 = (5k)^2 = 25k^2$

SO $5b^2 = 25k^2$

$\Leftrightarrow b^2 = 5k^2$

THEREFORE b^2 IS A MULTIPLE OF 5 AND
SO b IS A MULTIPLE OF 5

SINCE b IS AN INTEGER

IT HAS BEEN SHOWN THAT a AND b ARE MULTIPLES OF 5 SO THEY SHARE FACTOR OF 5

MAKE IT CLEAR WHERE THE CONTRADICTION IS

THIS IS A CONTRADICTION OF THE ASSUMPTION $\sqrt{5}$ IS RATIONAL, IT CANNOT BE WRITTEN AS $\frac{a}{b}$, SO, $\sqrt{5}$ IS IRRATIONAL

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